

helix-code-1.2.0-dev-0.0.1

HelixCode Release Changelog — helix-code-1.2.0-dev-0.0.1

Revision: 3 **Last modified:** 2026-08-09T12:38:49Z **Maintainer:** HelixCode release engineering (documentation draft, AI-agent authored; independently reviewed per §11.4.142/§11.4.194 — see “Independent review” section below)

DRAFT — pinned to a stated HEAD, this is not a cut tag. No git tag has been created by this document. Everything below was derived by reading real git history (`git log`, `git show`) — revision 1–2 content on 2026-07-28, the revision 3 content on 2026-08-09 at 12:38 UTC. Multiple background work streams were actively landing commits in this repo and its submodules while this draft was written (constitution §11.4.103 / §11.4.192 continuous multi-track work), so HEAD will have moved by the time this is read. **Re-derive the commit ranges below before treating any of this as final** — do not copy the SHAs or counts forward without checking.

Repo	Stated HEAD (revision 3)	Relative to its configured remotes
main (this repo)	5725ee068c965dabf2720ff6063acf3753a6c232	3 ahead, 0 behind on all four remotes (github/ gitlab/ origin/ upstream) — measured <i>after</i> a real git fetch --all --prune, not from stale

Repo	Stated HEAD (revision 3)	Relative to its configured remotes
		tracking refs (§11.4.6)
constitution (pin)	177f2b05545f2fe73fbb432ad0de3a6bed7edcf4	clean; carries §11.4.236/ §11.4.237/ §11.4.238
submodules/helix_llm (pin)	3d789cb888a09cfecdd2f0007a08a96d91caf5e9	clean; verified to contain 8260cf8 , the HXC-244 health fix, so a fresh clone gets it
submodules/helix_agent (pin)	66a3c1c6133a0ebdfeaa2ca4f1231588467cd145	working checkout is at d6856cc9 and differs from the recorded pin (git submodule status shows +); three other agents were actively editing this submodule and it was deliberately not touched

Repo	Stated HEAD (revision 3)	Relative to its configured remotes
		by revision 3
submodules/ tool_schema (pin)	8cec90baadad281eb26a5671b792921f03eb7386	clean

Revision 3 scope. Revisions 1–2 documented 5ab97d8c . . c29e1dcc. Revision 3 re-derives from c29e1dcc . . 5725ee06 — a further **231 commits** spanning 2026-07-28 → 2026-08-09, of which exactly **2** are the literal Auto-commit message and 229 are substantive; **0 merges** anywhere in the release range. Total since the last tag helix-code-1.1.0-dev-0.0.3 (5ab97d8c): **249 commits** (18 + 231, and git rev-list --count 5ab97d8c . . HEAD independently returns 249). The revision 1–2 content below is **preserved, not rewritten** — it was independently reviewed and its claims re-verified; revision 3 adds the new range, corrects one counting error revision 2 introduced (see “Independent review”), and refreshes every pin and remote position.

Format note: docs/changelogs/ did not exist in this repository before this draft — it was created now. The repository’s existing root CHANGELOG.md uses a ## [tag] – date per-release section with Added / Changed / Fixed / Known-gaps groups; this draft follows that convention but adds explicit short-SHA citations per entry and a per-repo commit inventory, so every claim traces back to a real, readable commit (constitution §11.4.6 — no invented history). Per the task scope, nothing here was committed with git add -A or similar — only this one file is staged, by explicit path.

Version

helix-code-1.2.0-dev-0.0.1 — draft, not yet tagged.

Prefix resolves to helix-code from .env HELIX_RELEASE_PREFIX (§11.4.151).

Prior tags in this lineage:

Tag	Date
helix-code-1.0.0-dev-0.0.1	2026-07-08
helix-code-1.1.0-dev-0.0.1	2026-07-09

Tag	Date
helix-code-1.1.0-dev-0.0.2	2026-07-13
helix-code-1.1.0-dev-0.0.3	2026-07-24

On the version jump: `helix-code-1.2.0-dev-0.0.1` is a **1.1.x → 1.2.0 minor bump**, not the next `-1.1.0-dev-0.0.4` patch in the existing sequence. This was named explicitly by the operator as this release's target tag; per instruction for this task, that is recorded here as an **operator decision**, not re-litigated or treated as unexplained drift.

Summary — what this release actually delivers for a user

- **Model identity is now honest end-to-end.** A client asking for an alias (e.g. `"model": "default"`) previously got that same alias echoed back in the response, on both the OpenAI-compatible and Anthropic-compatible wires, streaming and non-streaming. The response now names the model that actually served the request (e.g. the real `.gguf` path), matching `CONST-036/037`. Live SSE evidence was captured against a real running coder backend for both wires. One honest exception remains: Anthropic's streaming `message_start` event fires before any chunk exists, so it still carries the requested model — documented in-source, not silently left as a gap.
- **tool_schema translations now work regardless of where you run from.** Previously, calling into this library's `i18n` from any working directory other than its own module root silently returned raw translation keys instead of translated text (e.g. `toolschema_git_desc_commit` instead of “Create git commit”). Bundles are now compiled into the binary (`go:embed`), so this can no longer happen.
- **A large batch of false test failures in helix_agent is fixed.** A liveness check used to accept *any* HTTP response under 500 as “the server is up” — so when a different local service (LLMsVerifier) held the same default port, its 404 response was read as a healthy HelixAgent, and roughly a dozen test packages failed against the wrong service with nothing actually wrong in the product.
- **The whole platform can now be installed and booted as systemd units** (`./setup.sh`) — HelixCode server, HelixAgent, HelixLLM gateway + coder, infra (Postgres/Redis/Weaviate/Qdrant/ChromaDB/Cognee/Ollama), and now LLMsVerifier — with real HTTP health checks and a verified stop/start cycle, rather than requiring manual, host-specific setup.

- **LLMsVerifier is wired to real provider data and publishing real scores** instead of an empty {} — the gateway now routes on measured scores instead of always falling back to its static table.
- **go test ./... no longer breaks on generated challenge output** — a gitignored-but-still-scanned results directory could fail three otherwise unrelated packages depending on what an LLM happened to answer during a challenge run.
- **helix_agent boot no longer requires ChromaDB/Cognee on their compiled-in ports**, and no longer hangs indefinitely trying to start a vector store that is already running under a different owner (committed in helix_agent but see the pinning note below).

Fixed

SHA	Repo	Summary
e330149f	main	LLM response now reports the model that actually served the request (OpenAI + Anthropic wires, streaming + non-streaming), not the requested alias. Four layers: <code>llm.LLMResponse.Model</code> field + JSON round-trip, <code>OpenAICompatibleProvider</code> populating it from both non-stream and per-chunk stream responses, and both wire facades preferring the served model. Guards at <code>internal/server/wire_facade_model_identity_test.go</code> and <code>internal/llm/openai_compatible_stream_model_test.go</code> , each proven to fail when its fix is reverted.
66d6fb29	main	<code>go test ./...</code> no longer descends into <code>tests/e2e/challenges/test-results/</code> (renamed to <code>_test-results/</code> , the Go toolchain's own “not a package” convention) and tries to compile an LLM's prose answer as Go source. Unblocked 3 packages.
3268fd73	helix_agent (<i>pinned</i>)	<code>testutil.ServerAvailable</code> now requires a real <code>HelixAgent</code> -shaped health payload (HTTP 200 + identity check), not just “any status under 500”. Fixes a false-failure cluster of roughly a dozen packages caused by a different local service (LLMsVerifier) answering on the same default port. Three additional call sites that

SHA	Repo	Summary
		bypassed the shared helper entirely (a raw TCP dial, an error-only skip check) were also fixed.
0165ab1d	helix_agent (pinned)	Precondition gate's container-ownership check fixed in both directions: it no longer flags an unrelated third-party Postgres as "our" duplicate instance (false positive from {port , port+10000} guessing), and it no longer silently no-ops because it ran the wrong runtime command (false negative). Ownership is now established from the actual container runtime, filtered to this project's own containers.
0dfa69e1	helix_agent (pinned)	HelixAgent boot no longer hardcodes ChromaDB (:8001) / Cognee (:8000) endpoints — both resolve via HELIXAGENT_DEP_HOST / HELIXAGENT_PORT_CHROMADB / HELIXAGENT_PORT_COGNEE with the old literals as fallback. Vector-store startup now checks reachability first instead of unconditionally trying to become a second owner of an already-running store, which had been blocking the HTTP listener bind indefinitely.
8cec90b	tool_schema (pinned)	Translation bundles are go:embed-ed, so tr() lookups no longer depend on the process's current working directory. Previously any consumer calling in from outside this module's own root got raw key strings back instead of translations.
69a5ab8d	helix_agent (committed, NOT yet pinned — see Known gaps)	Fixes a goroutine leak in the Ollama provider's CompleteStream: 4 of 6 channel-send sites had no ctx.Done() escape hatch, so a caller that cancelled its context and stopped draining left the background goroutine parked forever. RED/GREEN regression guard added, driven via a real stuck-HTTP-response scenario and a live goroutine-stack inspection.
a345c551	helix_agent (committed, NOT yet	Two tests/unit test defects fixed (product behavior was correct in both cases): TestCompletionHandler_Models now wires a real model source instead of asserting on an intentionally-empty list;

SHA	Repo	Summary
	pinned — see Known gaps)	TestCompletionHandler_Stream now distinguishes an honest empty response (client-gone, request-timeout path) from an actual failure, instead of treating the httpptest recorder’s default 200 as an unconditional pass signal.
dd3c0c3b	main	Closes the third confirmed instance of an “unprioritized select” race this session: autoSaveLoop’s per-tick select between <- ticker.C and <- stop had no priority ordering, so a coincidentally-pending tick could still fire one more SaveAll() after DisableAutoSave() had already closed stop — the cause of TestAutoSaveLifecycle_ReEnableTicks’s “disabled auto-save must not tick” flake. Fixed by extracting the per-iteration select into autoSaveTick(tickerC, stop) with a non-blocking priority pre-check on stop. Full RED/GREEN/pre-fix/post-fix polarity evidence captured (§11.4.115): 2488/5000 forced-race hits pre-fix with the priority check disabled, 0/5000 post-fix. Found missing from this changelog’s original draft during independent review — added in revision 2; see “Independent review” section below.

Added

SHA	Repo	Summary
c4115c73	main	. /setup.sh now installs and boots the entire platform as systemd units (helix.target + 5 services) — HelixCode server, HelixAgent, HelixLLM gateway, HelixLLM coder, and infra — with a documented port map,

SHA	Repo	Summary
		boot-persistence via Linger=yes, and a verified full stop/start cycle. Fixes a KillMode=control-group bug that had been silently killing all ten infra containers' port forwarders whenever the infra unit's own health check failed.
d3e2c6e7	main	Adds LLMsVerifier as a 7th systemd unit, wired to the gateway. Implements the /api/scores endpoint upstream in the submodule (project-agnostic; consumer supplies the port via the unit). Gateway's log changed from "verifier unreachable" to actually reaching the verifier and correctly falling back on an empty score map.
98db7d79	main	LLMsVerifier is now seeded with a real, reachable provider (llamacpp → the local llama.cpp coder server) via a consumer-owned config/llmsverifier/config.yaml, instead of the submodule's own empty llms: [] default. First real verification score published (0.96) from genuine inference, replacing the previously-empty /api/scores.

SHA	Repo	Summary
410afda5	helix_agent (<i>pinned</i>)	Hyper provider discovery, registry, and verifier types.
1296d328	main	Adds scripts/ setup.sh scaffolding (157 lines) and two systemd unit files (helixcode-infra.service, helixllm-gateway.service) — precursor work later built out by c4115c73 / d3e2c6e7. Commit message is the literal Auto-commit; see Known gaps.

Changed

SHA	Repo	Summary
925aa859	main	Backfills the constitution's §11.4.182 TRUNK RULE (work on trunk is always labelled Track 1) across this project's five root governance carriers, and bumps the helix_agent and tool_schema submodule pins to the commits described above.
f14a3f89	main	Bumps the helix_agent submodule pointer (Hyper provider).
565506f6	main	Bumps HelixLLM to helixllm-0.2.0 (CPU inference).

SHA	Repo	Summary
c20215bd	helix_agent (<i>pinned</i>)	Chaos/compliance/automation test hardening carried in from prior sessions and committed so it is not lost — chaos/api, chaos/auth, module-compliance, config-generation automation, plus new helix_endpoint contract tests. go build ./... verified exit 0; the commit itself is explicit that the full suite was not independently reviewed or proven green at the time.
4b5f2867	helix_agent (<i>pinned</i>)	gofmt alignment and comment-spacing pass across the module.
36597718	helix_agent (<i>pinned</i>)	Search test reconciled to a reordered UnknownType check; adds coverage for the reachability short-circuit.

Infrastructure / Governance / QA evidence

SHA	Repo	Summary
578de68d	main	Lands §11.4.83 end-user QA evidence for 7 feature commits, captured against a freshly rebuilt binary (111 assertions, 0 FAIL, 0 SKIP). Explicitly documents that an earlier evidence pass had validated a 15-day-stale binary

SHA	Repo	Summary
		and was therefore invalid and re-run.
1c9fdcf2	main	§11.4.83 evidence against live paid providers (Cerebras, Cohere) and a real MCP wire for 3 further commits; drops the qa-evidence-gate violation count from 6 to 3.
e952f4d1	main	§11.4.83 evidence for the last 3 non-feature commits (CORS allowlist, Fyne desktop build docs, cognee determinism), each carrying an explicit “honest scope statement” about limited/no end-user surface. Drives the qa-evidence gate (G7) from 13 to 0 violations across 28 feature-shipping commits.
d8c44618	main	§11.4.83 evidence specifically for the e330149f model-identity fix (8/8 assertions PASS, mutation table proving each guard fails when deleted).
a5462e35	main	Live SSE proof, against the real coder backend, that <code>stream:true</code> reports the served model on every chunk (13 frames captured, 1 distinct model id, 0 frames carrying the alias).
c29e1dcc	main	Live proof for the Anthropic wire (POST /v1/messages returns the real served-model path) and for the tool_schema i18n fix (a compiled consumer binary resolves translations from unrelated working directories).
cf26a173	main	

SHA	Repo	Summary
		Bumps 4 owned submodule gitlinks (constitution, challenges, helix_qa, helix_agent) after pushing them to their upstreams. Commit itself states the full test sweep was not green at the time and no tag was cut.
0a4eb8d0	main	Corrects stale live-state anchors in RESUME.md after the above push — the exact “stale handoff doc” trap the project’s own governance warns about, caught and fixed in the same session.
56f7edf7	main	Message is the literal Auto-commit (no descriptive body). Diffstat shows real, substantive content: nil-request guards for the Replicate and Together LLM providers, a malformed-URL guard for the llamacpp provider, a security_scan tool rewrite, and an overhaul of verify_qa_evidence.sh / verify-governance-cascade.sh / obsolete_details_gate.sh / summary_sync_gate.sh, plus a helix_agent pointer bump. See Known gaps.

Revision 3 — the c29e1dcc..5725ee06 range (231 commits, 2026-07-28 → 2026-08-09)

Everything from here to “Per-repo commit inventory” was derived on 2026-08-09. Commit-type histogram over the 229 substantive commits: 87 docs, 74 fix, 27 chore, 15 test, 9 feat, 7 items, 3 track, 2 governance, 1 probe, 1 docs+test. 72 distinct HXC ids are named in commit subjects (HXC-151...HXC-248).

Summary — what changed for a user in this range

- **The gateway’s primary capability was dead and now works.** Every completion returned HTTP 500 “all providers exhausted”: with no cloud keys configured the local provider was the only completion path, and it was pointed at port 50052 — a default matching no service in the deployment, while the local model is served on 18434. Completions now return 200 with real model output.
- **The health endpoint no longer reports a verdict it never earned.** It returned `{"status": "healthy", "components": []}` and could return nothing else. It now names four checked dependencies, each with a measured round-trip duration.
- **You can now tell semantic search from a hash fallback.** The embeddings response previously carried no signal at all about which pipeline served it.
- **The desktop application no longer corrupts its own UI state under load** — 265 data races eliminated, root-caused to a *false* code comment claiming a widget setter was goroutine-safe when the toolkit’s own source says the opposite.
- **Streaming-with-tools no longer deadlocks** — a guaranteed (not racy) hang, plus a family of provider send-leaks that stranded a goroutine and an open HTTP body whenever a client disconnected.
- **Infrastructure “started” now means “serving”, not “created”** — the boot path previously reported success once containers existed, before anything answered.
- **Two credential exposures were found and handled**, one of which had been published on all four mirrors for 48 days.
- **Test suites that could not fail were made falsifiable** — 16 HelixQA banks covering 72 steps declared no assertions at all and scored PASS against a dead target; 238 steps now carry explicit expectations.

Fixed (revision 3 range)

SHA	Repo	Summary
15f00801	main	HXC-233 — completions were returning HTTP 500 “all providers exhausted” on every request. LocalRPCPort defaults to 50052 and the live process had the env var unset, so the gateway dialled a port no service listens on while <code>helixllm-coder.service</code> served the model on 18434. Fixed with two <code>Environment=</code> lines placed <i>before</i> <code>EnvironmentFile=</code> so an operator <code>.env</code> can still override. Verified: <code>:50052</code> connection refused → after fix, HTTPS 200 naming the

SHA	Repo	Summary
		real .gguf. Findable only because HXC-235's deploy an hour earlier replaced a misleading TLS error with an honest one.
fee7e1bc	main	HXC-233 guard — scripts/testing/guard_hxc233_completion_path_live.sh, asserting a <i>real</i> assistant message and a model id, not a status code. Its load-bearing proof is Mutation B: a stub returning HTTP 200 with an error envelope makes the guard exit 1, which a status-only assertion would have passed.
70cb8821	main	HXC-244 RED baseline — establishes as fact that cmd/helixllm/main.go:148 builds health.NewChecker() and registers nothing; of the 41 .Register() call sites in the tree, zero are on the health checker. The guard deliberately asserts the component list , not status=="healthy" — that assertion passed on the broken build for the wrong reason. Two defects in the guard itself were caught by its own golden-good/golden-bad fixtures (§11.4.107(10)): an invalid f-string made the analyzer crash on every input, and the crash's exit 1 read as “defect found”, so revision 1 reported RED-confirmed against a <i>healthy</i> fixture. Fixed structurally — analyzer findings start at exit 2, and any exit outside {0} ∪ {2..6} is a broken instrument and hard-fails in both polarities.
262a44ad	main	HXC-244 closed at the RUNTIME layer. The source fix (helix_llm 8260cf8, 969 insertions across 9 files with tests) had merged but never reached production: the running binary's mtime predated it and go tool nm \ grep -c registerHealthChecks returned 0. Closed by rebuild + redeploy of the gateway only — §9.2 backup first, artifact verified <i>before</i> install (registerHealthChecks 6, Ping values 4), NRestarts stayed 0. After: llm_providers, kv_cache_redis,

SHA	Repo	Summary
		vector_store_qdrant, llms_verifier, each carrying a measured duration — the evidence of a real round-trip that distinguishes this from a hardcoded list.
f2e12570	main	HXC-235 deployed — the deployed binary was 15 days stale , containing semantic_embeddings zero times, so four helix_llm commits (HXC-233's fix among them) were not live either. The process-start time had suggested a 13-hour gap; strings on /proc/<pid>/exe gave the real answer in one command. Post-deploy the key appears with value false, which is <i>correct</i> (HELIX_EMBEDDING_PROVIDER is unset, so the pipeline genuinely is the HashEmbedder) — the defect was that a caller could not tell.
724d2bb0	main	HXC-229 — the gateway ran in its web framework's debug mode under systemd. Environment=GIN_MODE=release added to the unit; verified on the running process via /proc/<pid>/environ, with 0 debug lines in a window provably containing a real startup.
e879702c	main	265 data races eliminated in applications/desktop. Root cause: a code comment asserting (*widget.Entry).SetText is goroutine-safe, which the toolkit's own thread.go has contradicted verbatim since v2.6. RED 265 WARNING: DATA RACE exit 1 → GREEN 0 races exit 0 at -count=3.
905a0b0a	main	A guaranteed StreamWithTools deadlock (not a race): GenerateStream was inlined at both call sites writing into a 100-buffered channel nobody drained. Plus six provider files with unguarded sends leaking a goroutine and an open HTTP body on client disconnect.
dd3c0c3b	main	Third confirmed “unprioritized select” race — autoSaveLoop could fire one more SaveAll() after DisableAutoSave() had closed stop. 2488/5000 forced-race hits prefix, 0/5000 post-fix. (<i>Documented in revision 2</i>

SHA	Repo	Summary
		<i>but filed under the wrong range — see “Independent review”.)</i>
0bcd2e98 + 6d8a4920	main	HXC-228 — infra “started” now means “serving” rather than “created”; verified by a full-target cold boot with zero restarts.
11861996	main	HXC-168 / CONST-042 — a database password carried as a literal in shipped setup and container files published to all four mirrors. Reachability was <i>established, not assumed</i> : two compose files published Postgres on all interfaces, and the autoboot one boots by default. Explicitly does not close the item — rotation is outstanding.
be599c10 + ce5ecc04	main	HXC-167 — transcript redaction was fail-open and allowlist-only; three further fail-opens found, one of them <i>inside the guard itself</i> .
e6118f99	main	Dropped inverted docs_chain derive edges that would have destroyed 156 tracked items .

Added / Changed (revision 3 range)

SHA	Repo	Summary
ae218ee3	main	Cascades §11.4.236 (QA-deploy-readiness gate — no manual-QA hand-off until the mandated validation produced a candidate-fingerprinted PASS verdict; a blocker must bind to a seam, never prose) and §11.4.237 (mandatory exhaustive context-and-spirit-aware translation review — accuracy necessary, never sufficient) into all six consumer carriers, closing a measured §11.4.157 lockstep gap (canon at .237, consumers at .235). Verified byte-identical across all six (md5 df2cb867668b82e66e2d25235cd1dec5).
74f24528	main	Cascades §11.4.238 (automated QA must be the DISCOVERER: manual QA finds

SHA	Repo	Summary
		nothing new, and anything it finds is a coverage escape) into all six carriers and bumps the constitution pin to 177f2b0 in the same commit per CONST-049 step 7, after verifying the head published on all 8 constitution remotes. §11.4.238's forensic anchor is <i>this project</i> : every defect closed that day — the health endpoint, the dead completion port, the two unfetchable pins, both hardware-profiler defects — was found by an agent reading source or probing by hand, while HelixQA reported green throughout.
f0dd7069	main	Bumps four submodule pointers so a fresh clone actually gets the fixes — every fix lived in a submodule while the parent still pinned a commit predating it. Every old pointer verified an ancestor of its new head (FF-safe ×4, no force), every head verified published. Notable §11.4.6 datum: <code>git rev-list --count HEAD..@{u}</code> reported behind=0 <i>before</i> the fetch; after <code>fetch --all</code> , behind=3 — remote state is not knowable without fetching.
204bc7b4	main	Two third-party pins recorded on main that no clone can fetch . The dispatching premise was wrong and the correction is the finding: 7 of 9 suspected pins were fine. The real defect is that signoz and skyvern were explicitly listed as deliberately-not-recorded, then recorded anyway by an Auto-commit. Two independent oracles agree neither exists upstream (GitHub REST 422; live upload-pack “not our ref”).
33ee0670	main	HXC-243 — of 129 HelixQA banks / 243 http steps, 16 banks / 72 steps declared no assertions at all and the executor compares a field only when declared, so they scored PASS on any response.

SHA	Repo	Summary
		Baseline measured: 16/16 banks PASS against a dead target, exit 0. Now 238 steps carry an explicit expectation, 5 an explicit skip+reason, 0 assertion-free. Key finding: expect_status alone would not have fixed it — the wrong port answers 200 with a JSON-RPC error envelope.
d8e46001	main	A HelixQA agent proved a PASS-bluff in our own test banks by construction — it pointed a bank at the wrong service (a completion path already proven dead with HTTP 503) and wrote its thesis before running. Both cases reported PASS.
9f8fdd27	main	HXC-227 — a live provider key published in a tracked design doc, committed 2026-06-19 and present on all four mirrors for 48 days , under a heading reading “already configured”. The discovering agent’s handling is recorded as correct: it recognised that exporting the doc would copy the key into two <i>new</i> tracked files, removed the artifacts and quarantined the source rather than attempting a scrub. Only rotation withdraws it.
09a086a6	main	HXC-187 / D-7 — renames the thin root module to dev.helix.code/meta so the root and inner modules stop claiming the same identity.
1dab3e06 + 40c8d5ce	main	Tune test parallelism from the binding cgroup quota , not nproc. Paired with 918f969c, which explicitly corrects its own earlier root cause (“GOMAXPROCS does not explain the agent stalls”).
732320c8	main	Untracks 104 MB of vendored node_modules from the agent repository.
	main	

SHA	Repo	Summary
ee39ea30, 9f68ed04, cf4eb22a, e9a1ecb6, 69772e56		§11.4.65 sibling-export backfill (101 root docs, 73 inner docs, 62 under docs/**, script companions, 4 missing .pdf). These account for the bulk of the range's 458k inserted lines.

Tracker / workable-items (revision 3 range)

SHA	Repo	Summary
ac9e9e4d, 38d8f641	main	Closed verified items in two passes; High severity 8 → 4, Critical 0. 38d8f641 also caught that the shipped workable-items binary was ten days older than the invariant its own source defines , so validate reported OK on records the source rejects.
8214bd93, e84e5b4c	main	HXC-217 — 15 closure records described their proof in prose instead of pointing at it, which is mechanically indistinguishable from having no proof. Narratives preserved verbatim in a new <code>item_history.note</code> column; a resolvability validator added that refuses a closure whose stated location does not resolve.
679e2e2e + 825e1c3c	main	HXC-245 filed, then its own filed root cause retracted as wrong, refuted by the code —

SHA	Repo	Summary
		the correction is the artifact worth reading.
79939789	main	HXC-246 — the agent test suite reports 74 failures, filed with classification <i>deliberately open</i> . Three signatures, none yet proven a product defect and none yet cleared: 22 test files hardcode a cache port that is closed on this host (the container publishes it elsewhere); a cluster failing in 0.00s while the service under test is provably healthy; wall-clock assertions overshooting ~26% under measured contention. The commit withdraws one of its own guesses explicitly.
b48d235e	main	HXC-247 — two owned, individually-correct components both claim port 8100. The agent's port allocator assigns it 8100 while its shipped config still pins 7061, so the verifier takes 8100 first; 82 test files address 8100 expecting the agent, receive the verifier's 404, and report the agent broken while it is healthy. Root cause of HXC-246's 0.00s cluster. Contains a self-correction retracting an earlier "8090" claim with measurements.

SHA	Repo	Summary
d706f8cc	main	HXC-248 — integration-test cleanup decides whether to tear down the stack by checking only that <i>something</i> answers on port 8100. That something is the verifier, not the agent, so teardown is skipped and the live platform survives by coincidence ; stop or move the verifier and the same check would tear down a running deployment. The log line it prints is also false.
1f177495	main	HXC-244 filed — the defect was found, fixed, deployed and verified, but no tracker entry was ever created; HXC-243 and HXC-245 existed and 244 was a genuine gap. Filed directly as Fixed (→ Fixed.md) against the runtime evidence above.
7bd80ff6	main	Swept all 48 non-terminal items against <code>git log --all -i --grep=<ID></code> ; closed HXC-199 and HXC-217 on re-verified evidence and deliberately left 46 open. Records a systemic finding: <code>verify-all-constitution-rules.sh</code> invokes 21 <code>scripts/gates/</code> entries and zero of the three <code>scripts/testing/</code>

SHA	Repo	Summary
		guard_hxc*.sh guards written this cycle — a guard nothing runs cannot fail (§11.4.135).
5725ee06	main	Normalises 16 severity values that held entire triage paragraphs where a closed-set token belongs. 10 were safe to reduce because the rationale already rendered from body_md; 6 were not — the column was their only copy, so the rationale was first written into body_md and only then reduced. Prose 16 → 0, valid 230 → 246, with all 16 rationales proven to survive a full regeneration.

Per-repo commit inventory

main (this repo) — 18 commits in 5ab97d8c . . c29e1dcc (revisions 1–2 range)

*(Revision 3 correction. Revision 2 changed this figure from 18 to 19 on the grounds that dd3c0c3b had been omitted. The omission was real; the count change was not. git rev-list --count 5ab97d8c . . c29e1dcc returns **18**, and git merge-base --is-ancestor c29e1dcc dd3c0c3b succeeds — dd3c0c3b is dated 2026-07-28 18:13 and lies **after** c29e1dcc, i.e. in the revision 3 range, not this one. Revision 2 correctly added the missing commit to the Fixed table but attributed it to the wrong range and inflated this count to match. Restored to the measured 18; dd3c0c3b is carried in the revision 3 range above. The two figures now reconcile: 18 + 231 = 249 = git rev-list --count 5ab97d8c . . HEAD.)*

```
dd3c0c3b fix(persistence): close unprioritized-select race in autoSaveLoop
c29e1dcc docs(qa): LIVE proof for the Anthropic wire and the i18n cwd fix
a5462e35 docs(qa): LIVE proof that stream:true reports the served model
e952f4d1 docs(qa): §11.4.83 evidence for the three non-feature commits – G
```

1c9fdcf2 docs(qa): §11.4.83 evidence for the Cerebras, Cohere and MCP-conc
d8c44618 docs(qa): §11.4.83 evidence for e330149f model-identity fix
925aa859 docs(governance): backfill the §11.4.182 TRUNK RULE + bump two su
e330149f fix(llm/server): report the model that ACTUALLY served the reques
578de68d docs(qa): land §11.4.83 end-user QA evidence for 7 feature commit
0a4eb8d0 docs(resume): correct live-state anchors after the end-of-session
cf26a173 chore(submodules): bump gitlinks after end-of-session sync
56f7edf7 Auto-commit
66d6fb29 fix(e2e): stop go test compiling generated challenge output (3 pa
98db7d79 feat(llmsverifier): seed real providers via consumer-owned config
d3e2c6e7 feat(systemd): add LLMsVerifier as the seventh unit, wired to the
c4115c73 feat(systemd): install and boot the whole platform – HelixCode +
1296d328 Auto-commit
565506f6 chore: bump HelixLLM to helixllm-0.2.0 (CPU inference)
f14a3f89 chore: bump HelixAgent (Hyper provider)

submodules/helix_agent — 9 commits since the previously-pinned 55ad06c6

Pinned into main's current gitlink (3268fd73) — 7 commits:

3268fd73 fix(testutil/tests): verify the server is HelixAgent, not just th
0165ab1d fix(precondition): establish container ownership from the runtime
c20215bd test(helix_agent): chaos/compliance/automation hardening + live-e
4b5f2867 style: gofmt alignment and comment spacing across helix_agent
36597718 test(search): reconcile UnknownType assertion to the reordered ty
0dfa69e1 fix(boot): resolve dependency endpoints from env; don't block sta
410afda5 feat: Hyper provider discovery + registry + verifier types

Committed in helix_agent's own history but NOT yet reflected in main's gitlink — 2 commits (see Known gaps):

69a5ab8d fix(ollama): close goroutine-leak in CompleteStream on ctx cancel
a345c551 fix(tests/unit): wire the model source; stop reading recorder-def

submodules/tool_schema — 1 commit since the previously-pinned cb7bf675, fully pinned

8cec90b fix(i18n): embed translation bundles so lookups work from any dire

Known gaps / not yet verified

This section exists so this document does not read as more finished than the underlying work is. None of these were papered over.

1. **§11.4.185 manual QA-team confirmation has not been given.** No record of a manual QA sign-off for this scope exists anywhere in this repository. Per this project's own governance, automated green is necessary but not sufficient — a manual QA pass is required before this scope can be called fully complete, and it has not happened.
2. **Desktop GUI packages do not build on this host.** Directly attempted `go build ./applications/...` in `helix_code/` on this host and it fails with two real, host-environment errors, not code defects:
 - `github.com/go-gl/gl/v2.1/gl: pkg-config cannot find gl.pc.`
 - `github.com/go-gl/glfw/v3.3/glfw: fatal error: X11/Xlib.h: No such file or directory.` This is not a new discovery — a prior commit in this cycle (`d6c05f76`, already merged before this release's commit range) documented this same gap and verified the README's claimed error text is byte-identical to the real one on this host. Re-confirmed here, not re-litigated. Any package depending on the Fyne desktop GUI stack is affected.
3. **The zero-skip full integration test stack was not booted for the last sweep.** Checked directly (`podman ps`): the containers currently running on this host (`helixcode-infra-postgres`, `-redis`, `-weaviate`, `-qdrant`, `-chromadb`, `-cognee`, `-ollama`, etc., all `systemd`-managed via `compose.helixcode-infra.yml`) are the **production-like platform infra**, not the `docker-compose.full-test.yml` zero-skip test stack (which would run containers named `helixcode-postgres-full`, `helixcode-redis-full`, etc. — none of those are present). Any integration-test result gathered without first running `make test-infra-up` for that specific stack is environmental, not a confirmed pass or fail, until it is booted and the sweep re-run against it.
4. **The full go test suite has not concluded cleanly on a quiet host for this cycle.** `docs/CONTINUATION.md`'s own rev-18 audit (not independently re-run as part of this documentation-only task, to avoid adding more load to a host other agents are actively building/testing on) records one completed `helix_agent` full-suite run on a contended host reporting 18 FAIL packages, and attributes this to ambient host contention (constitution §11.4.119) consistent with this project's own documented pattern — but that specific 18-package list has not itself been re-run on a quiet host and confirmed clean. Treat as **unconfirmed**, not cleared.

5. **helix_agent's goroutine-leak fix (69a5ab8d) and test fix (a345c551) are real, committed work but are NOT yet part of this release's pinned artifact.** Main's current gitlink for submodules/helix_agent points at 3268fd73; the submodule's own HEAD is two commits further, at 69a5ab8d. Until the gitlink is bumped and re-verified, this release does not actually ship the Ollama goroutine-leak fix even though the fix exists and is committed.
6. **helix_agent's working tree is dirty** (modified tracked files and untracked new test files observed at read time) and **2 commits ahead of its own configured remotes** — i.e. not yet pushed anywhere outside this host. This draft does not push anything.
7. **main is 8 commits ahead of every configured remote (0 behind)**, i.e. also not yet pushed. This document commits only itself; per this task's constraints, nothing is pushed and no tag is created.
8. **Two commits carry the literal message Auto-commit** (56f7edf7, 1296d328) with no descriptive body. Real, substantive content was recovered from their diffstats (see the Infrastructure / Added tables above) rather than invented, but the commit messages themselves do not describe intent, verification, or evidence — a documentation-clarity gap at the commit-message layer, flagged rather than silently backfilled.
9. **No claim in this document was accepted from a stale project artifact without independent re-derivation of the specific git facts it depends on** (HEAD SHAs, gitlink pins, tag list, container names) — those were each re-checked directly against the live repositories at draft time. The one exception is item 4 above, which is explicitly attributed to docs/CONTINUATION.md rather than re-run, and is labeled unconfirmed for exactly that reason.
10. **helix_agent has moved further since this draft's stated pin (69a5ab8d), adding a third sibling to the same defect class this document already tracks.** Commit e2f42b12 (fix(llm,database): prioritize cancellation in racy select statements) fixes exactly the two "sibling" locations dd3c0c3b's commit message names as instances of the same unprioritized-select pattern — internal/llm/lazy_provider.go's createProviderWithContext and internal/database/debate_log_repository.go's StartCleanupWorker — with the same RED/GREEN polarity-evidence discipline (§11.4.115). It is real, committed work, but as of this revision it is **not yet part of this release's pinned artifact**: main's gitlink for submodules/helix_agent still points at 3268fd73, four commits behind helix_agent's own current HEAD (3268fd73 → a345c551 → 69a5ab8d → e2f42b12). This compounds Known gap 5 rather than replacing it — do not treat 69a5ab8d as the current

unpinned tip; re-derive submodules/helix_agent's actual HEAD before citing it, per this document's own opening warning.

Known gaps added or re-measured in revision 3

Gaps 1–10 above were written against the revision 1–2 range. Re-checked on 2026-08-09; the following supersede or extend them.

1. **Three standing guards written this cycle are registered in no runner.** Measured: `scripts/verify-all-constitution-rules.sh` invokes **21** `scripts/gates/entries` and **0** `scripts/testing/guard_hxc*.sh`. There is no auto-discovery glob. So `guard_hxc229_gateway_release_mode.sh`, `guard_hxc233_completion_path_live.sh` and `guard_hxc244_health_components_registered.sh` all exist, are individually falsifiable, and **never execute**. §11.4.135 asks for a standing suite; these are standing files. This is the direct reason HXC-229 was left open rather than closed in 7bd80ff6.
2. **db-to-md silently drops two items.** `workable-items validate` reports HXC-247 and HXC-248 have no `doc_segments` row, and a regeneration to a scratch path confirms it empirically: both appear **0** times in the regenerated `Issues.md`, while every other item appears. 2 of 442 items. Pre-existing — reproduced identically against a pre-change backup — and deliberately **not** silently repaired, because synthesising the missing segments is adjacent to HXC-245's failure mode and deserves its own item.
3. **13 open items carry no severity, so §11.4.132 risk-ordering cannot rank them.** 196 rows have an empty severity, but 183 of those are already closed, where severity ranks nothing. The actionable set is the 13 open ones: HXC-227, 229, 231, 232, 233, 234, 235, 236, 239, 240, 241, 242, 243. No rule exists in this repository to derive a severity — the schema itself says *"informational only; not closed-set"*, with no CHECK and no default — so these were left alone rather than invented (§11.4.6).
4. **submodules/helix_agent's recorded pin and its working checkout disagree.** `git submodule status` shows a + prefix: the pin is 66a3c1c6, the checkout is d6856cc9. Three other agents were actively editing that submodule while revision 3 was written, so it was not entered and the delta is **not characterised here**. HXC-234 could not be resolved past ambiguous for the same reason — its fix lives inside that submodule.
5. **The release position is 3 ahead / 0 behind on all four remotes,** measured *after* a real `git fetch --all --prune`. Supersedes gap 7's "8 ahead". The three local commits are revision 3's own tracker work (1f177495, 7bd80ff6, 5725ee06). Nothing was pushed and no tag was cut.

6. **Two further Auto-commit commits** landed in the revision 3 range (c209bff2, 29696107), extending gap 8. Additionally the pinned helix_11m head 3d789cb itself carries the literal message Auto-commit — it was verified by content (it contains 8260cf8), not by its subject.
7. **§11.4.185 manual QA-team confirmation is still not given** — unchanged from gap 1, and now load-bearing for §11.4.238, which this range cascaded: manual QA is a confirmation step that must find nothing new, and it has not been run at all for this scope.

Independent review

Per constitution §11.4.142 (every change gets an independent review, no exception) and §11.4.194 (exhaustive all-scenario review), this document received a full independent review after its original commit (8bdac5b0) landed. The authoring agent had been killed by an API session limit immediately after committing and never reported, so this changelog had zero review until now.

Method: every cited short SHA in the Fixed / Added / Changed / Infrastructure tables and the per-repo commit inventories was checked to exist and to actually do what its entry claims, via `git show --stat` and full `git show` against this repository, submodules/helix_agent, and submodules/tool_schema — not accepted on the strength of the subject line alone. Quantitative claims (assertion counts, PASS/FAIL tallies, live scores, frame counts) were spot-checked against the cited commits' own bodies. `git log --oneline 8bdac5b0..HEAD` was run in this repo (empty — main has not moved) and, using the changelog's own stated submodule pins as the base, in submodules/helix_agent (one further commit, e2f42b12 — see Known gap 10) and submodules/tool_schema (none). The `.env` HELIX_RELEASE_PREFIX resolution, the §11.4.44 revision header, and the “3 desktop packages fail to build” claim (applications/desktop, applications/aurora_os, applications/harmony_os — confirmed all three import the Fyne stack, so all three are genuinely covered by `go build ./applications/...` and by the stated X11/OpenGL failure) were all independently re-verified rather than taken on trust.

Finding (blocking, now fixed in revision 2): commit dd3c0c3b — this changelog's own direct parent commit, committed roughly 90 seconds before 8bdac5b0 — was completely absent from every section of the original draft despite being real, substantive, in-scope work covered by the document's own stated data-collection window. It has been added to the Fixed table, the per-repo commit inventory, and cross-referenced against its helix_agent sibling fix (Known gap 10).

Verdict: no invented history was found anywhere in this document — every other cited SHA, quantitative claim, and Known-gaps disclosure checked out exactly as

stated. With the one finding above corrected, this document is independently confirmed accurate as of that revision (2026-07-28T15:21:59Z).

Revision 3 (2026-08-09)

Why it was needed: the document was pinned to a HEAD 11 days and 231 commits stale, and covered none of the work in that range.

Method: the range was re-derived from `git log / git show`, never from memory or from any prior project artifact. Every short SHA newly cited above was checked to exist and to carry the subject attributed to it — **51 main-repo SHAs verified present via `git cat-file -e`, 0 missing**, plus `helix_llm 3d789cb` and `8260cf8` and `constitution 177f2b0` in their own repositories. The two `helix_agent` values (`66a3c1c6`, `d6856cc9`) are quoted from main-repo data only — `git ls-tree HEAD` and `git submodule status` — precisely because that submodule was not entered. Counts are pasted command output, not estimates: 231 commits in `c29e1dcc..HEAD`, 2 of them the literal Auto-commit, 0 merges, 249 since the tag. Remote positions were measured **after** an actual `git fetch --all --prune` rather than from tracking refs, because this range contains a commit (`f0dd7069`) whose own message documents a tracking ref reporting `behind=0` before a fetch and `behind=3` after.

Finding 1 (corrected here): revision 2 introduced a counting error while fixing a real omission. It raised “commits since the last tag” from 18 to 19 to account for `dd3c0c3b`. `git rev-list --count 5ab97d8c..c29e1dcc` returns 18, and `dd3c0c3b` is provably *after* `c29e1dcc` — it belongs to the revision 3 range. The commit did need adding to the Fixed table; the count did not need changing. Both are now correct and reconcile: $18 + 231 = 249$.

Finding 2: a claim in the revision 1–2 pin table could not have been true as written and is superseded. It described submodules/`helix_agent` as pinned at `69a5ab8d`. The pin recorded in HEAD’s tree today is `66a3c1c6`, and the working checkout is a third value, `d6856cc9`. Rather than reconcile a stale figure, revision 3 states all three positions and marks the delta uncharacterised (Known gap 14), because the submodule was deliberately not entered.

Verified rather than assumed: that the pinned `helix_llm (3d789cb)` actually contains `8260cf8`, the HXC-244 health fix — `git merge-base --is-ancestor` confirms it, so a fresh clone gets the fix. This mattered because this same range contains two separate incidents (`f0dd7069`, `204bc7b4`) of pins that did not carry the fixes attributed to them.

Honest boundary (§11.4.6): revision 3 has itself had **no** independent review — it is authored by the same agent that produced the tracker commits it describes. Its factual claims are individually re-derivable from the commands cited, but the

§11.4.142 independent-reviewer seam has not been crossed for this revision. Treat the revision 3 sections as verified-by-construction, not peer-reviewed.

Sources verified

All SHAs, commit subjects, commit bodies, tag names/dates, gitlink pins, container names, and the go build error text above were read directly from this repository, submodules/helix_agent, and submodules/tool_schema via git log, git show, git tag, git ls-tree, podman ps, and a direct go build invocation on 2026-07-28. No claim in the Fixed / Added / Changed tables above is unsourced from a real commit; anything that could not be traced to a commit was left out rather than inferred.

Revision 3 (2026-08-09). The c29e1dcc..5725ee06 range was derived with git rev-list --count, git log --oneline/--format, git show --stat, git log -1 --format=%B on every commit cited, git merge-base --is-ancestor for the ancestry claims, git ls-tree HEAD <path> and git submodule status for the pins, git fetch --all --prune followed by git rev-list --count refs/remotes/<r>/main..HEAD for the remote positions, git ls-files to confirm cited evidence directories are actually **tracked** rather than merely present on disk, and sqlite3 -readonly plus workable-items validate / sync db-to-md for every tracker figure. The constitution anchors were read from constitution/Constitution.md at the pinned commit and confirmed to be genuine ### block-openers, not incidental mentions. File sizes were taken with wc -c, not stat, which misreports on this host. Nothing in the revision 3 sections is carried over from a prior document without re-derivation.